Chapter 14
Power Device Induced Oscillations and Electromagnetic Disturbances

14.1 Frequency Range of Electromagnetic Disturbances

Every power electronic switching action results in a deviation from the ideal sinusoidal AC current or the ideal homogeneous DC current. Switching events are usually done periodically in time. Every periodic event can be separated into a row of sinus and cosinus terms by means of Fourier transformation. With this tool, the generated frequencies, the harmonics and their intensity can be calculated.

Figure 14.1 shows a rough overview of the disturbances and oscillations caused by power electronics. It is distinguished between disturbances created by switching events in power-electronic converters, the harmonics of the switching frequency on the low- and medium-frequency range, and the device-induced high-frequency disturbances.

At low frequencies, e.g. in phase-commutated converters, the disturbances caused by the input rectifier occur as multiples of the grid frequency of 50 – 60 Hz, their intensity declines proportionally to 1/n. In self-commutated converters with modern power devices, the typical switching frequencies for converters using IGBTs are in the range of 1 – 20 kHz; also in this case, harmonics of the respective switching frequency are to be expected. Higher switching frequencies can be applied with MOSFETs as power devices. In switch-mode power supplies, nowadays frequencies up to 1 MHz and above have become possible.

Device-induced oscillations, on the other hand, result from switching events. Since the switching times of the devices are much smaller than the period of the switching frequency, the electromagnetic disturbances caused thereby occur in a frequency range significantly higher than the switching frequency.

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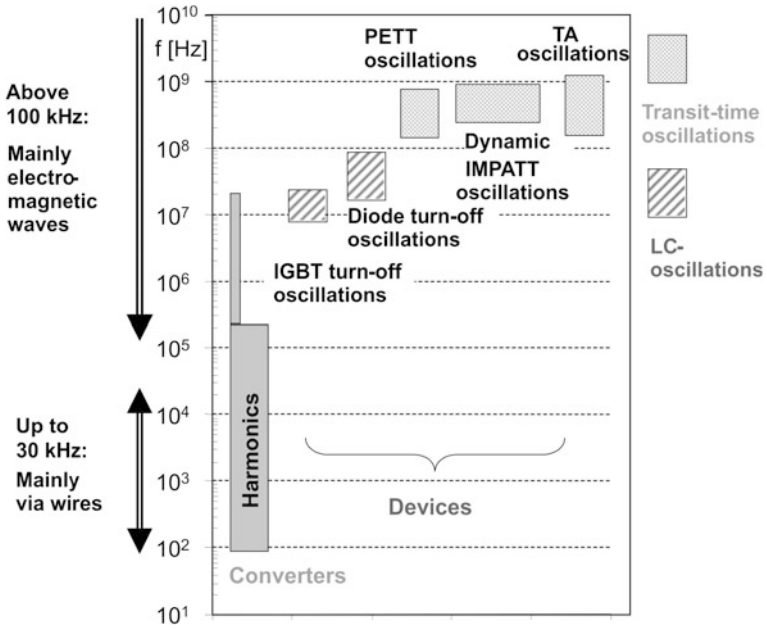


Fig. 14.1 Typical frequency ranges of disturbances caused by power-electronic effects

Electromagnetic disturbances of a frequency < 30 kHz spread mainly via wiring and cables. They disturb the electric grid in the form of grid feedback. Electromagnetic disturbances of frequencies > 100 kHz are radiated off to a large extent as electromagnetic waves, and this can cause incompatibility with other electronic and power-electronic equipment.

Harmonics

Figure 14.2 shows two simplified examples of electric signals, either depicting the current or the voltage curve.

For a rectangular course with point symmetry to π and the amplitude a, as shown in Fig. 14.2a, the harmonics can be calculated using the Fourier transformation

$$y = \frac{4a}{\pi}\left(\sin\,\omega t + \frac{1}{3}\sin\,3\omega t + \frac{1}{5}\sin\,5\omega t + \cdots\right) \tag{14.1}$$

Multiples of the switching frequency $f = \omega/2\pi$ are generated, i.e. the 3rd, 5th, 7th, … harmonics. With increasing ordinal number n, their intensity decreases proportionally to 1/n.

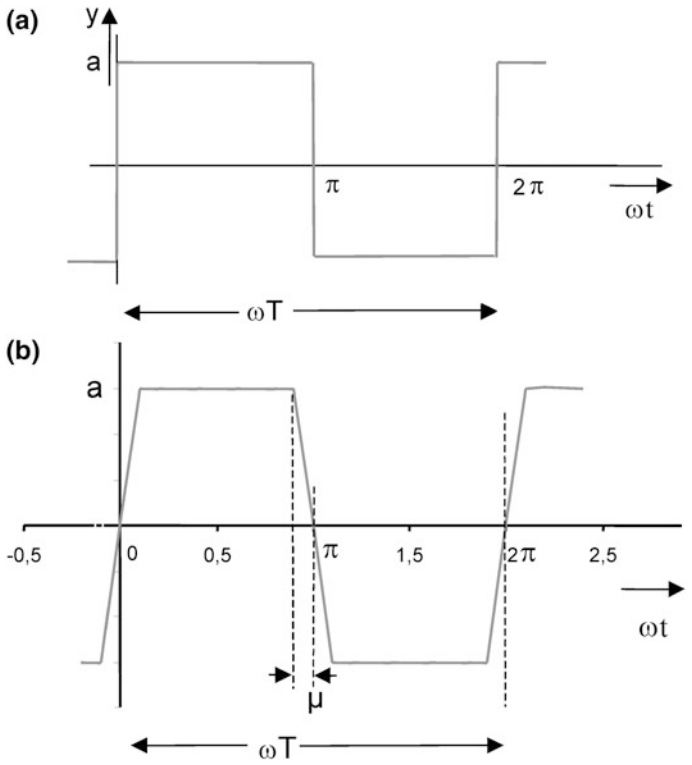


Fig. 14.2 **a** Rectangular course of a signal. **b** Trapezoidal course of a signal

However, for a trapezoidal course with point symmetry to π and the amplitude a as shown in Fig. 14.2b, the Fourier transformation results in

$$y = \frac{4}{\pi}\frac{a}{\mu}\left(\sin\,\mu\,\sin\,\omega t + \frac{1}{3^2}\sin\,3\mu\,\sin\,3\omega t + \frac{1}{5^2}\sin\,5\mu\,\sin\,5\omega t + \cdots\right) \quad (14.2)$$

The amplitudes of the harmonics decrease proportionally to $1/n^2$. For a non-symmetric shape, additional terms will occur. Nevertheless, the faster decline of the harmonics and therefore a trapezoidal course is much more suitable. The slopes *di*/*dt* – represented in Fig. 14.2 with μ – can be adjusted by gate resistors if MOSFETs and IGBTs are used. To reduce harmonics, the switching times are reduced by increased gate resistors. However, this will increase switching losses. A trade-off must be made in many applications between switching losses on the one hand and electromagnetic emissions on the other hand.

Suitable filters are implemented as a countermeasure to improve the electromagnetic compatibility. This shall not be explained further in this chapter; the attention is riveted instead on oscillations created by power devices themselves.

14.2 LC Oscillations

14.2.1 Turn-off Oscillations with IGBTs Connected in Parallel

In power modules, often a lot of single dies are connected in parallel. It is very difficult to give all single dies identical symmetrical conditions in respect to the length of the current-conducting path to the main terminals as well as in respect to the length of the wiring of the drive signals. Thermal conditions must be considered, too. Often trade-offs must be made. Figure 11.33 shows an example of parallel connection of five IGBT dies, in which asymmetric tracks for the main current to the respective die are given. The wires for the drive signals, for which also a symmetrical setup is of importance, are not drawn in the schematic circuit diagram of 11.33b.

Figure 14.3 shows the measurement of a parallel connection of two IGBTs. To create differences, the gate resistor are chosen slightly deviating: for chip 1, a resistor of 6.02 Ω, and for chip 2 of 6.45 Ω [Pal99]. The IGBT chips rated to 100 A are exposed to a forward current of 20 A in each chip. According to the lower gate resistor, chip 1 starts with the turn-off process. This has the consequence that the current in chip 2 increases in the first instance, while the total current is still at the same level. Then, during the current decline, an oscillation between the two chips builds up. A period of 50 ns can be read, which corresponds to a frequency of 20 MHz. The oscillations cannot be seen in the course of the total current in this case. Only the measurement of the single currents shows the oscillations occurring between the chips.

Figure 14.4a shows turn-off oscillations in a first version of a press-pack IGBT housing, see also Fig. 11.4. The current in single chips of the many chips connected in parallel is measured. The current shows a high-frequency oscillation in the range of 10 MHz.

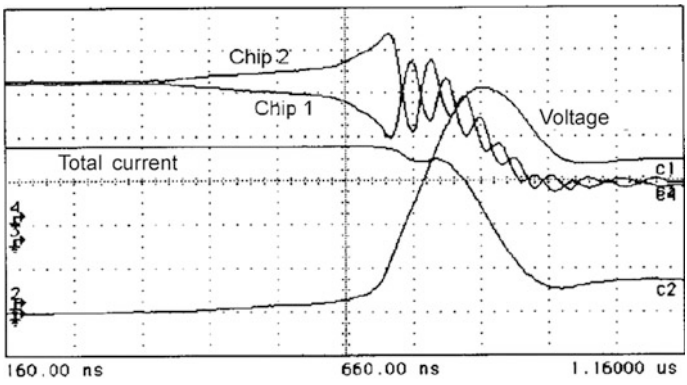


Fig. 14.3 Oscillations of the current in two IGBTs connected in parallel. Gate resistors 6.02 Ω for chip 1 and 6.45 Ω for chip 2. Current 10 A/div, voltage 50 V/div. Figure according to [Pal99] © 1999 EPE

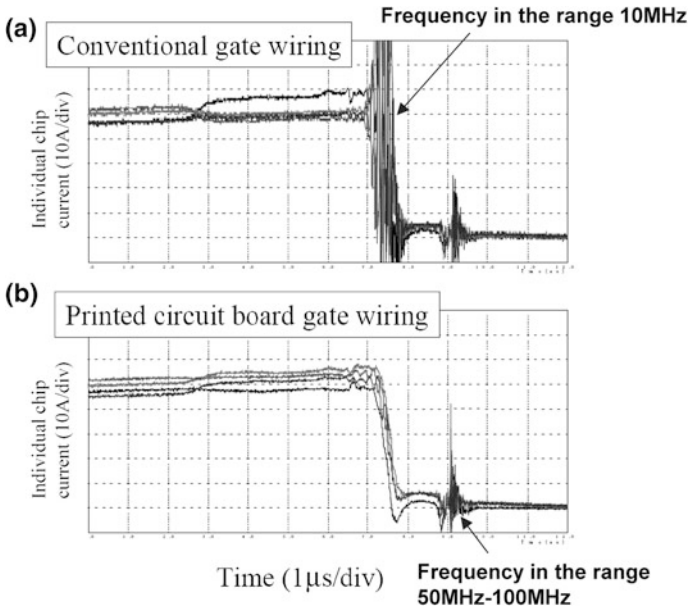


Fig. 14.4 **(a)** Turn-off oscillations in a press-pack IGBT and **(b)** their elimination by a symmetrical arrangement of the gate signal connections with an integrated PCB. Figure from [Omu03] © 2003 IEEE

If the control terminals are realized with a printed circuit board (PCB) which leads to the same conditions for each of the 42 chips connected in parallel, the turn-off oscillations can be eliminated, as can be seen in Fig. 14.4b. Such a PCB, shown in Fig. 14.5, has copper layers on both sides; one side is the potential of the gate signal, the other side is the potential of the control emitter terminal. Very close to every single chip, gate resistors are arranged on the PCB.

The frequency range in which turn-off oscillations are found is in the range between 10 and 20 MHz. This is clearly above the values which are to be expected for the harmonics of switching events (compare Fig. 14.1). Turn-off oscillations must be avoided not only because of electromagnetic emissions. They can additionally increase the turn-off losses of chips and can lead to thermal failures.

Possible countermeasures against turn-off oscillations are:

- The setup of an arrangement as symmetrical as possible. This is also to be considered for parallel connection of discrete devices as well as single modules.
- If this is not possible because of mechanical and other reasons, the gate resistors R_G of IGBTs and MOSFETs can be increased. This counteracts oscillations, but in the same way it increases the turn-off losses. For IGBTs, compare Eq. (10.6) and the discussion of the influence of gate resistors on example of MOSFETs in context with Fig. 9.19.

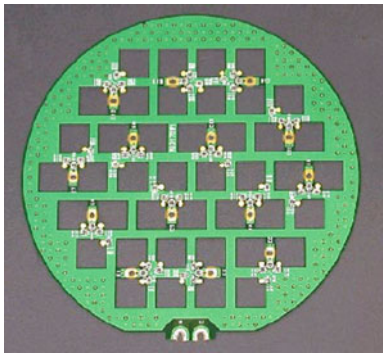


Fig. 14.5 PCB to ensure symmetrical control terminals in a press-pack IGBT. Figure from [Omu03] © 2003 IEEE

14.2.2 Turn-off Oscillations with Snappy Diodes

Fast diodes with insufficient reverse-recovery behavior more often cause oscillations in power circuits than asymmetries in IGBTs connected in parallel. For details on snappy recovery behavior see Chap. 5. Figure 14.6 shows the course of the voltage at turn-off of a snappy diode in a circuit according to Fig. 5.19. The application is a step-down converter in a battery-fed electric vehicle. Instead of the IGBT in Fig. 5.19, a 100-V rated MOSFET is used as power switch. The snap-off of the reverse current in the diode leads to a voltage peak of 100 V. The abrupt snap-off leads to a current overshoot in forward direction, the diode is turned off again, a second and a third voltage peak are generated, and finally the effect ends up in a damped LC oscillation.

The frequency of the LC oscillation generated by snappy diodes is determined by the device capacity C_j and the parasitic inductance L_{par}

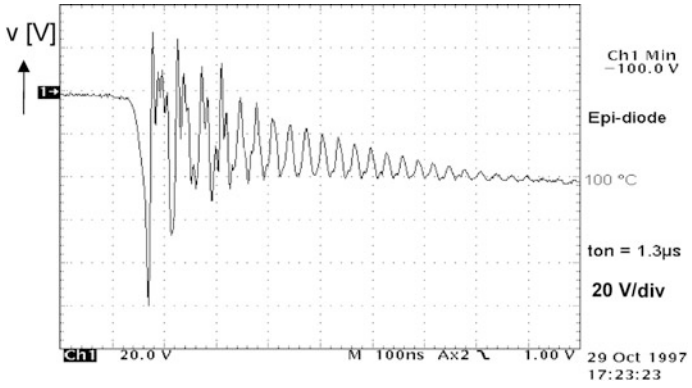


Fig. 14.6 Course of the voltage at turn-off of a snappy diode. Period 30 ns, frequency 33 MHz

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{L_{par} \cdot C_j}} \tag{14.3}$$

The equivalent electrical circuit for such an oscillator is shown in Fig. 14.7 [Kas97]. It must be noted that the capacity $C_j = c_j \cdot A$ is dependent on the voltage, see Eq. (3.113). The resistance R_b of the interconnections of the diode acts as damping component; the same holds for $R_{n,p}$ which stands for the base of the diode. During the turn-off process, electrons and holes are removed from the base and $R_{n,p}$ is not linear; in fact, it is strongly varying.

During the LC oscillation, C_j is not constant. For a diode rated to a blocking voltage of 1200 V with a steep doping profile of the p-layer at the pn-junction, C_j can be assumed to be 250 pF/cm^2 for the estimation of the expected frequency of oscillations [Kas97]. Considering the area of the diodes and some typical housings and their respective parasitic inductance, the values given in Table 17 are obtained Table 14.1.

For a diode rated to 100 A and 1200 V in an IGBT module of older construction type with a typical parasitic inductance of 100 nH, the frequency of LC oscillations caused by snappy diodes is expected to be in the range of 48 MHz. With a module of modern architecture, approximately 20 nH are given as parasitic inductance; the frequency moves to a range of 100 MHz. In a high-power module rated to 1200 V, 12 diodes rated 100 A are connected in parallel and a much larger capacity is given. Because of the larger volume of the module, longer interconnections are necessary. Despite of this, the parasitic inductance in such modules has been reduced significantly. Depending on the inductance, frequencies in the range of 5–15 MHz are to be expected for the discussed oscillations.

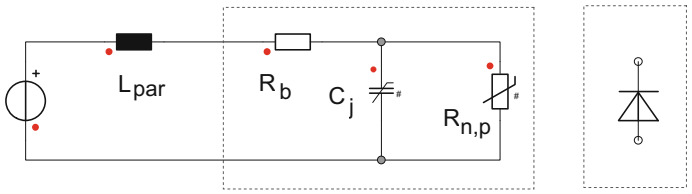


Fig. 14.7 Equivalent circuit for an oscillator consisting of diode and its parasitic components. Figure from [Kas97]

Table 14.1 Estimation of the frequency range of LC oscillations for 1200-V free-wheeling diodes

	C_j	L_{par} (nH)	f (MHz)	$T = 1/f$ (ns)
Bipolar 100-A diode, active area 0.44 cm^2	110 pF	20	107	9.3
		100	48	20.8
1200-A module, diodes	1.32 nF	100	13.9	72
		800	4.9	204

If soft recovery diodes are used, the described oscillations can be avoided. Soft recovery diodes are producible today and they are indispensable for the operation of a modern power-electronic converter.

However, it must be noted that not every oscillation in conjunction with diodes is originated by snappy turn-off behavior. If a non-suited unsymmetrical parallel connection is done, oscillations may occur even if soft-recovery diodes are used. An example is given in Fig. 14.8 [Eld98]. In this example, three diodes are connected in parallel; the diode D1 is located close to the main terminals, D2 and D3 with additional wiring are connected in parallel. The diode D1 with the lowest inductance is commutated with the highest *di/dt*. The reverse-recovery current maximum is first reached in D1. The process continues in D2, and then in D3, with an increased commutation velocity di/dt. It is maximal for D3, because the reverse current is already declining in D1 and D2. During the reverse-recovery process, the current oscillates between the devices. Finally, at the end of commutation, these internal oscillations superimpose to an oscillation of the total current i_D, too.

If a parallel connection is given and oscillations in the reverse recovery of free-wheeling diodes are found, it also has to be investigated whether this is caused by asymmetrical arrangement of the wirings and interconnections.

Oscillations have even been found with a single soft-recovery diode, if the condition is fulfilled that the switching time of the diode t_{rr} agrees with half the period of the resonance of an LC oscillator circuit. In applications with MOSFETs and IGBTs, this can be verified easily: the modification of the gate-resistor R_G of the used IGBT or MOSFET can vary the turn-on slope of the transistor and with it the commutation velocity *di/dt*. So the switching time t_{rr} of the diode is modified, and in this case the oscillations should vanish.

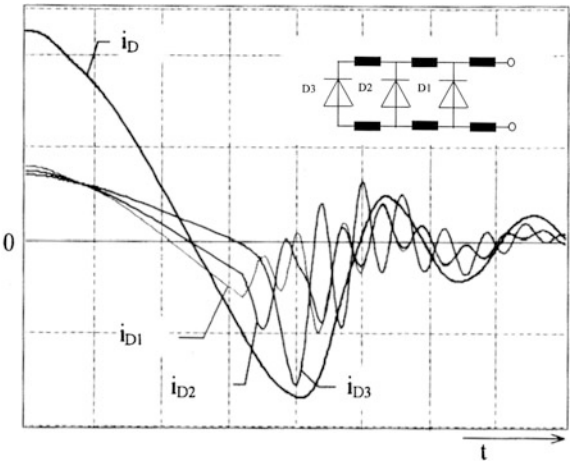


Fig. 14.8 Course of the current during reverse recovery in a parallel connection of diodes with different wiring inductances. 50 ns/div, 50 A/div, 25 °C, V_{bat} approx. 300 V. Figure from [Eld98]

14.2.3 Turn-off Oscillations with Wide Bandgap Devices

If just Si is replaced by SiC in high-current applications, huge problems with oscillations may occur. This is reported for 300 A/1200 V SKiM power module application for a 70 kW inverter with 30 kHz switching frequency [Win12], where Si bipolar freewheeling diodes were replaced with SiC Schottky diodes.

The oscillation is an LC-oscillation between the junction capacity of the Schottky-Diode and the parasitic inductance, where C_j is

$$C_j = \sqrt{\frac{q \cdot \varepsilon \cdot N_D}{2 \cdot (V_j + V_r)}} \cdot A \tag{14.4}$$

and has its maximum $C_j(0\ \mathrm{V})$ when the applied voltage V_r is zero. The value $C_j(0\ \mathrm{V})$ is typically given in the data sheet. Due to the two decades increased N_D^+ with SiC for the same voltage (Fig. 3.19), C_j will be one decade higher in the case of the same active area. The current peak $I_{D,peak}$ caused by the capacity of the Schottky-diode is given by

$$I_D,peak = C_j(0V) \cdot \frac{dv}{dt} \tag{14.5}$$

with dv/dt as voltage slope created by the IGBT. As can be deduced from Fig. 14.9, dv/dt is about 30 V/ns. The Schottky diode used for the measurement in Fig. 14.9 had a capacity $C_j(0\ \mathrm{V})$ of about 754 pF which gives a current peak $I_{D,\ peak}$ of about 90 A, this agrees well with the measured result.

To reduce the oscillations in Fig. 14.9, the gate resistor of the IGBT had to be increased. This leads to increased IGBT turn-on losses. Finally, all advantages of the low-loss SiC diodes were lost if the requirements for low electromagnetic emission are to be fulfilled. Note, that the used SKIM module was already developed for low parasitic inductance. However, for the SiC Schottky diode application it is not sufficient. This shows that high current SiC diode applications are only possible if more effort is taken in new low-inductive packages.

A tail current, which occurs in Si bipolar devices and which damps oscillations, is missing with SiC. This is a challenge for a new packaging technology. Applying wide bandgap devices results in a strong need for low parasitics in the package. An example for acceptable switching behavior of SiC-MOSFETs in a low inductive housing is shown in Fig. 14.10.

The device package also contains, besides the known chip internal capacities C_{gs}, C_{gd} and $C_{ds,}$ the capacities formed by the substrate insulator layers, e.g. Al_2O_3 or AlN, which are displayed in Fig. 14.11b as $C_{\sigma+}$, $C_{\sigma out}$ and $C_{\sigma-}$. Another point to consider is that $C_{\sigma out}$ is recharged with every switching event, and an undesired current is supplied into the heat sink. If the two L_σ and the $C_{\sigma+}$, and $C_{\sigma-}$ are unbalanced, they also generate a current into the heat sink [Fei15].

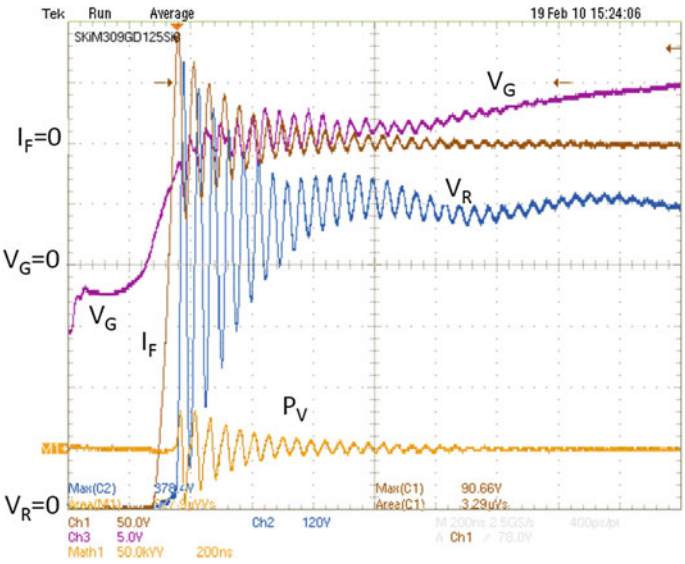


Fig. 14.9 Oscillations at turn-on of an IGBT with SiC Schottky diodes as freewheeling diode. V_R Voltage over the Schottky diode 120 V/div, I_F inverted diode current 50 A/div, V_G gate voltage of the IGBT 5 V/div, 125 °C. Figure from [Win12]

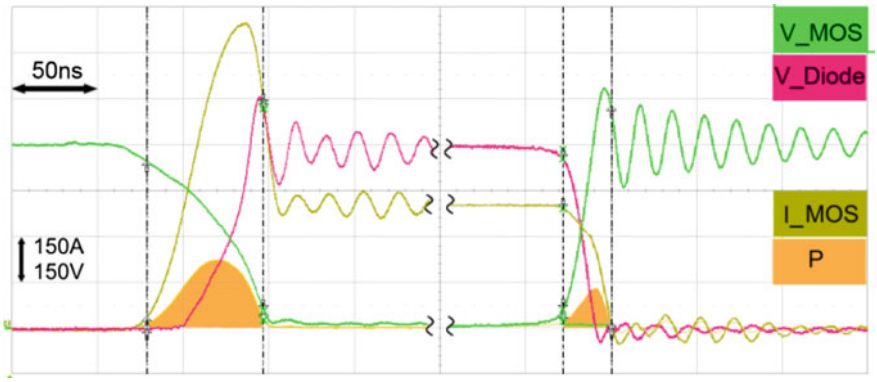


Fig. 14.10 SiC MOSFET with body diodes, 400A 600 V 150 °C dv/dt 40 V/ns. Figure from [Bec16]

Especially the parasitic inductance L has to be very low in a package for SiC and GaN. GaN as well, with the capability of high dv/dt and di/dt, needs low inductive packaging. There is much effort invested today in the development of packages with very low inductance. Special care has to be taken in case of paralleling, since non-symmetric inductivities in front of parallel connected chips will lead to bad dynamic current sharing and internal LC-oscillations, see Fig. 14.8.

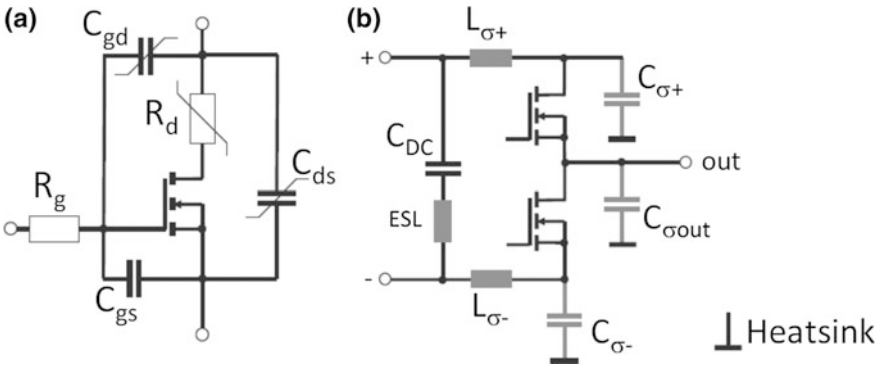


Fig. 14.11 Parasitic elements of a SiC MOSFET **(a)** In the chip **(b)** Switching cell parasitics. Figures from [Fei15] PCIM Europe 2015

14.3 Transit-Time Oscillations

A power device has a lowly doped middle layer with a thickness w_B At turn-off of a bipolar device, the existing charge carriers are removed; a part of them at an instant at which a space charge has already built up. The charge carriers flow through the space charge with at a drift velocity v_{sat} This results in a transit time for which a first approximation can be given with

$$t_T = \frac{w}{v_{sat}} \tag{14.6}$$

The drift velocities for electrons and holes $v_{sat(n)}$ and $v_{sat(p)}$ are given in Eq. (2.38), the width of the space charge w during switching processes is smaller or equal to the base with w_B. The carrier transit time corresponds to a frequency of $1/t_T$. Dependent on the base width, transit-time oscillations occur with a frequency in this range; it is the range between 100 MHz and 1 GHz or even higher, see Fig. 14.1. The occurring frequency depends on the type of effect and on its phase relations. This is shown in the following.

Transit-time oscillations are used for the fabrication of microwave devices which are used as microwave oscillators [Sze81]. With power devices, such oscillations must be avoided since they are a danger for the power device itself, and since their electromagnetic emission can cause unwanted effects in driver circuits and other electronic components of the adjacent environment. The occurrence of transit-time oscillations was observed for three effects in power devices. They are described in the following paragraphs, and measures to avoid them are discussed.

14.3.1 Plasma-Extraction Transit-Time (PETT) Oscillations

PETT oscillations can occur in a bipolar device at turn-off during the tail current interval. They have been observed with IGBTs as well as with soft-recovery free-wheeling diodes [Gut01, Gut02]. Figure 14.12 shows an example with IGBTs.

The oscillations are observed in the course of the gate voltage, but they occur primarily as oscillations of the collector current I_C and collector voltage V_C where they cannot be resolved easily because of their low amplitude. Therefore, the stray effect into the gate signal is mainly observed. The oscillations occur after the device has taken up the voltage and the device is in the interval in which a tail current flows.

An example of the occurrence of PETT oscillations in a soft-recovery free-wheeling diode is shown in Fig. 14.13 [Sie03]. The measurement was executed in a test setup of an IGBT module rated to 600 A and 1200 V. At turn-on of the IGBT, the free-wheeling diode is turned off, the PETT oscillations occur in the tail current interval of the free-wheeling diode. Only soft-recovery diodes can lead to PETT oscillations. However, soft recovery is essential for fast diodes in such applications. Since the resolution of the current measurement of the anode current of the diode is too low, the PETT oscillation is detected in this example with a wire loop which works as antenna and which is placed close to the diode.

The mechanism of oscillation is related to the mechanism of the Barrier Injection Transit Time (BARITT) diode which is intended as microwave oscillator. The BARITT diode has a metal-semiconductor-metal structure or a pn^-p-structure. At a voltage applied in blocking direction, the electric field reaches the opposite metal-semiconductor junction or the opposite p-layer, and it releases an injection of carriers there [Sze81]. In contrast to the BARITT effect, with the PETT effect the space charge reaches the remaining charge carrier hill or remaining plasma zone which is still a reservoir of free carries and feeds the tail current.

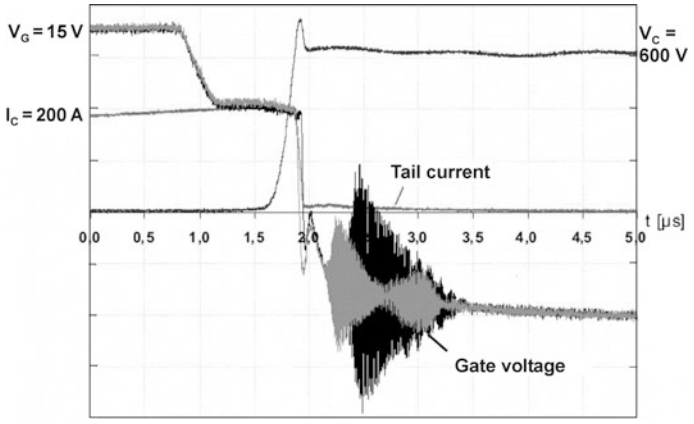


Fig. 14.12 PETT oscillations at turn-off of an IGBT. Figure from [Gut01] © 2001 IEEE

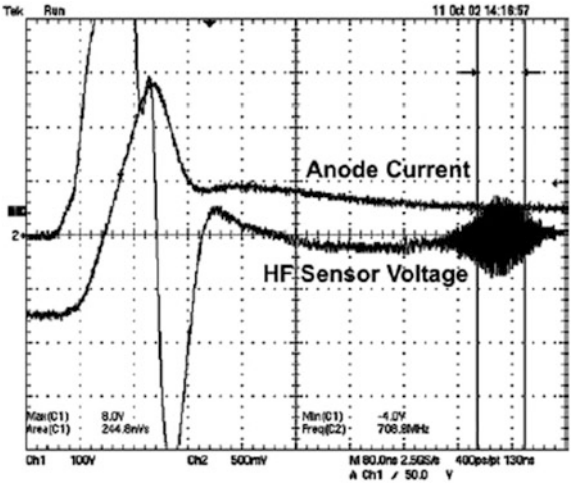


Fig. 14.13 PETT oscillation in the tail current of a soft-recovery free-wheeling diode at turn-on of an IGBT. V_{bat} = 600 V, I_F = 200 A, di/dt = 4000 A/µs, T = 300 K. Figure from [Sie03] © 2003 EPE

Figure 14.14 shows the process in the device at a PETT oscillation on example of a freewheeling diode. At the anode side (Fig. 14.14, right-hand side), an electric field has built up during the turn-off event; it takes up the voltage.

The charge carrier hill which feeds the tail current is still located at the cathode side (Fig. 14.14, left-hand side), compare Fig. 5.25 or 5.33. The tail current flows through the space charge as hole current. The shape of the electric field is triangular in this interval.

The occurrence of oscillations under such conditions is discussed in detail in [Eis98]. For a PETT oscillation, this is shown in Fig. 14.15. A high-frequency AC voltage $V_{RF}\cdot\sin\omega t$ superimposed to the DC link voltage V_{DC} is assumed (Fig. 14.15a). In the same way as in the BARITT effect, an injected j_{inj} current is generated as the AC voltage has its maximum at $\omega t = \pi/2$ (Fig. 14.15b). This injected current flows through the space charge region with the velocity v_d. The corresponding current density at the terminals of the device j_{inf} is expressed by the

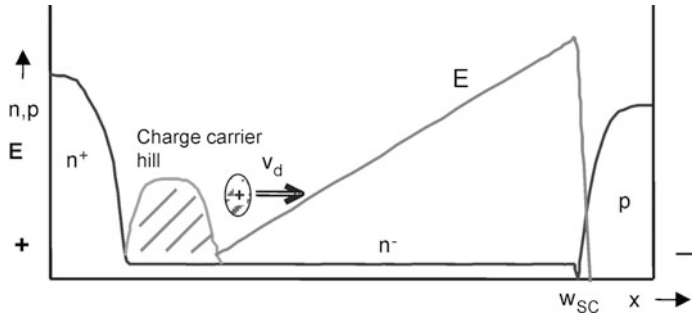


Fig. 14.14 Process in a freewheeling diode at occurrence of a PETT oscillation

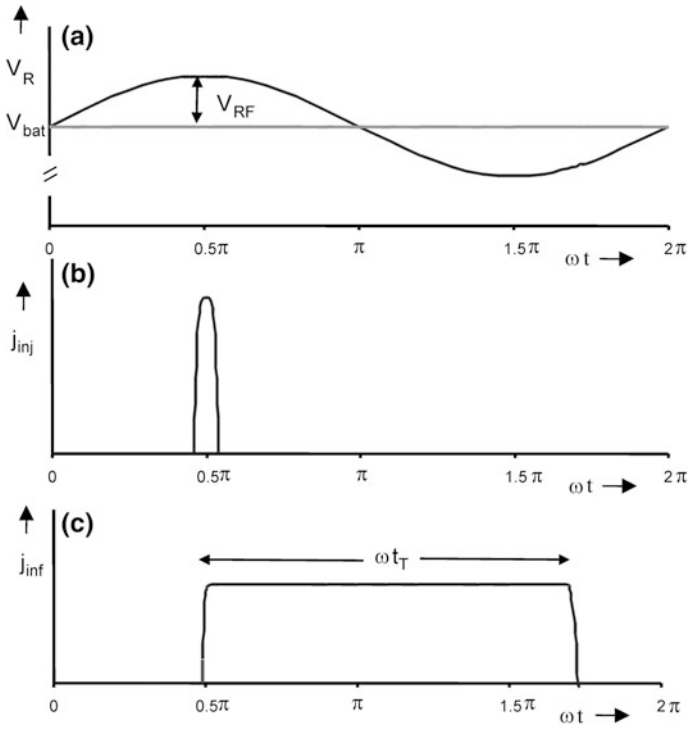


Fig. 14.15 Origin of PETT oscillations: **(a)** High-frequency AC voltage superimposed to the applied DC voltage **(b)** Injection current at $w = w_P$ at the time $\omega t = \pi/2$. **(c)** Current at device terminals. Figure from [Sie06b] © 2006 IEEE

Ramo-Shockley-theorem [Eis98] and shown in Fig. 14.15c. The current flow at the device terminals starts at $\omega t = \pi/2$ and is found in the time interval ωt_T, needed by the carriers to transit the space charge region. The generated RF power is given by

$$P_{RF} = \frac{A}{2\pi} \int_0^{2\pi} j_{\text{inf}}(\omega t) \cdot V_{RF} \sin \omega t d\omega t \tag{14.7}$$

In Figs. 14.15a and c one can see: P_{RF} is positive for $\omega t_T < \pi$, zero at $\omega t_T = \pi$ and negative at $\omega t_T > \pi$. A negative value of P_{RF} means that RF power is generated. The created RF power is maximal for $\omega t_T = 3\pi/2$ and decreases again for $\omega t_T > 3\pi/2$. As long as P_{RF} is of negative sign, the device acts as current source and emits *RF* power.

The transit time t_T is given by [Gut02]:

$$t_T = \int_{w_p}^{w_{SC}} \frac{1}{v_d(w)} dw \tag{14.8}$$

The velocity v_d in the space charge region depends on the strength of the electric field, compare Chap. 2, Eq. (2.39) and Fig. 2.15. The electric field is of a triangular shape in the given case [Sie06b] (see Fig. 14.11).

For the BARITT diode, the transit time is given in first-order approximation [Sze81] by Eq. (14.6) as $t_T = w_{SC}/v_{sat}$, where v_{sat} is the saturation drift velocity of holes under high-field conditions (approximately 10^7 cm/s in silicon). Note that the drift velocity v_d is lower than the saturation velocity v_{sat} for a significant part of the space charge region. Continuing with this simplification and taking into account the point of maximum RF-power generation at $\omega t_T = 3\pi/2$, the frequency of the PETT oscillations is approximated by

$$f_T = \frac{3 \cdot v_{sat}}{4 \cdot w_{sc}} \tag{14.9}$$

as it is given also in [Sze81].

From Fig. 14.15 it follows that excitation of the superimposed AC power is possible in a specific frequency range. It can be concluded that PETT oscillations occur only when the "negative-resistance" behavior found during one period is greater than all other resistive components in the complete circuit. Moreover, the phase shift between the oscillation voltage and the AC current at the device terminals is essential for an occurrence of this kind of oscillation. Furthermore, it can be concluded from Fig. 14.15 that the efficiency of the RF generation is low, since power is always dissipated during the interval $\pi/2$. Low efficiency is also a characteristic of BARITT diodes [Sze81]. Therefore, PETT oscillations only occur if there is a resonance circuit formed by the junction capacitance and the inductance of the bond wires close to the device, whose resonance frequency has to be close to the frequency f_T according to Eq. (14.9).

Additionally, PETT oscillations cannot occur as long as there is a high amount of remaining plasma in the device and the corresponding reverse current is high, since this stored charge acts as damping which hinders the occurrence of the oscillations. Further important parameters are the applied voltage V_{bat}, since this voltage determines w_{SC}, and the temperature, since the drift velocity is temperature-dependent. It is typical for PETT oscillations that they only emerge at very special conditions. If one deviates from these conditions, no more PETT oscillations can be found. Therefore, the possible occurrence of PETT oscillations is easily overlooked in the procedure of the qualification of a power module.

Simulations of the PETT effect have been done already in [Gut02] and [Sie06b]. Detailed simulations in [Fuj12] point out that the velocity of a packet of holes is not constant but varies with the electric field strength in the drift region.

PETT oscillations are radiated-off as electromagnetic waves. This radiation can lead to the effect that the power-electronic equipment violates the requirements of electromagnetic compatibility (EMC). The EMC requirements are fixed in different standards, e.g. by the European standard EN55011 (international standard IEC CISPR 11) [DIN00]. For details see the special literature in this field.

Figure 14.16a, b show an experiment of measuring the electromagnetic radiation of the 600-A 1200-V module, from which the oscillations are shown in Fig. 14.12 [Sie03]. Since no chamber shielded against external electromagnetic radiation was available, first a measurement of existing electromagnetic disturbance of the environment was necessary. Figure 14.16a shows the result of the environmental EMC measurement which lasted 2 h, followed immediately by the subsequent measurements. The different grey-colored bars shown in Fig. 14.16a mark frequency ranges used by broadcasting and telecommunication. Obviously, the largest interfering signals were caused by mobile communication equipment. This

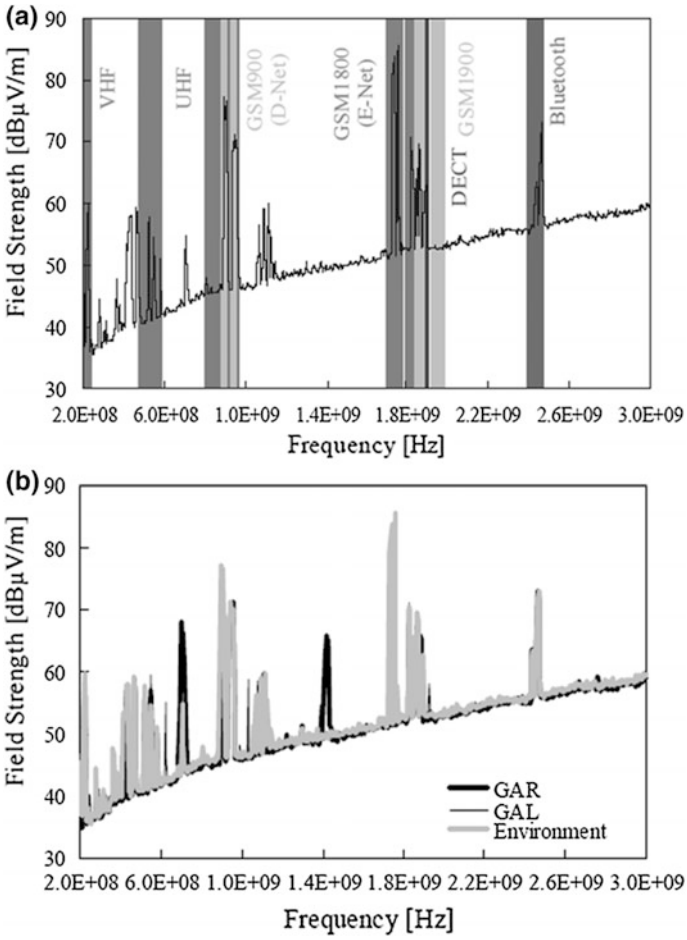


Fig. 14.16 EMC measurement of a PETT oscillation in comparison with environmental background radiation **(a)** Environment without operation of the module **(b)** with operation of the module PETT oscillations at the module GAR at 700 MHz and 1.4 GHz. Figure from [Sie06b] © 2006 IEEE

measurement series is included in Fig. 14.16b to give evidence of additionally generated signals that were caused by the PETT oscillations.

The PETT oscillation during the turn-off (marked as GAR) caused two sharp peaks in the frequency spectrum, appearing at 700 MHz and 1.4 GHz, which could be assigned to the fundamental frequency and the second harmonic.

The width of the space charge of the diode used in the module GAR amounts to approx. 85 µm for the given conditions. If the value of 8×10^6 cm/s is used as drift velocity for holes, a frequency of approx. 700 MHz results from Eq. (14.9), which agrees with the measured value.

Although the spurious radiation caused by PETT oscillation was relatively low in strength, exceeding the EMC limits could easily occur. In particular, this would be expected if more than one power module is used, as is typical for power-electronic equipment, and the radiations of the single modules are summed up.

In another application, PETT oscillations were found in a 1.8 MW high-frequency converter with an operating frequency of around 100 kHz. The setup of the equipment consisted of more than 100 power modules. The onset of the oscillation generated an error signal in the control unit [Sie06b]. Therefore, PETT oscillations must be avoided.

To prevent PETT oscillations, it is therefore not particularly helpful to modify the semiconductor device itself. Every power semiconductor has a space charge if voltage is applied, and it has therefore the capability for oscillations according to Eq. (14.9). However, it is essential to avoid an LC circuit that is in resonance with the transit frequency given in Eq. (14.9).

Figure 14.17 shows the setup of the module GAR, for which PETT oscillations in Figs. 14.12 and 14.17b were found. The oscillations occurred at the diode which was used as free-wheeling diode. It is marked as FWD in Fig. 14.17, left-hand side; in the equivalent circuit in Fig. 14.17 it is the diode at the bottom.

Figure 14.18 shows the impedance of this module on the left-hand side [Sie06b]. The three-dimensional EMC simulator FLO/EMC [FLO04] was used for the calculation. It solves the complete Maxwell equations numerically. The geometry of the module in Fig. 14.17 is used; the space is divided into cells modeled as the intersection of orthogonal transmission lines. There is no possibility to introduce real semiconductors into FLO/EMC. Therefore, a simplified model was used which reproduces the correct junction capacitance or on-state resistance of the devices (IGBT and FWD). For the characterization of the power modules, the excitation in the form of a delta pulse was applied across a FWD. In this way, the electric and magnetic fields and the resulting impedance can be calculated.

Figure 14.18 shows the simulation results for the impedance of the power module GAR (as seen from the FWD where the excitation was applied). A resonance point at a frequency of about 700 MHz resulted, which was in accordance with the oscillation frequency as given by the transit frequency f_T (Eq. 14.9) of the FWD. This resonance point is a necessary condition for the appearance of PETT oscillation, and 3D EMC simulation can be used to predict resonance points of a complex mechanical construction.

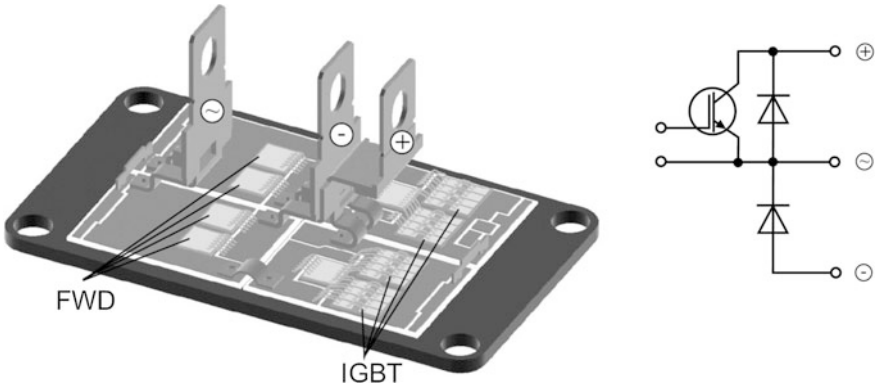


Fig. 14.17 Assembly of the module GAR which showed the PETT oscillations in the measurement in Fig. 14.12, and its internal electrical circuit. Figure from [Sie06b] © 2006 IEEE

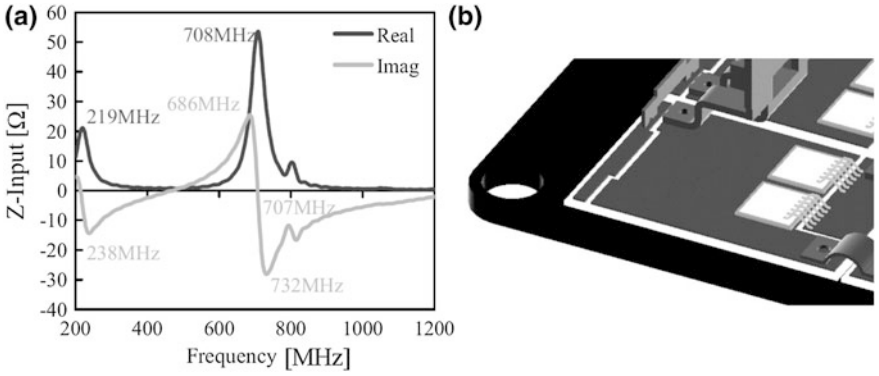


Fig. 14.18 Impedance of the module in Fig. 14.17 **(a)** Detail of the arrangement of the free-wheeling diodes at which the PETT oscillation occurs **(b)** Figure from [Sie06b] © 2006 IEEE

However, it must be mentioned that already small modifications in the assembly may shift the occurrence of PETT oscillations to other conditions, or may weaken it or even eliminate it.

The arrangement of the bond wires in the module GAR is shown in Fig. 14.18b. Considering the given chip area and the resulting space charge capacity, only a small inductance is possible for an LC circuit with resonance in the range of 700 MHz. It can be formed only by components in the immediate neighborhood of the chip, e.g. the next conductor paths on the DCB substrate, the bond wires and their arrangement.

An obvious and efficient way for lowering inductance would be to provide additional shorts between the anode contact areas, as published in an older patent application of 1995 [Zim95] and shown in Fig. 14.19b. This results in a clear suppression of the module resonance in the transit-frequency range, as shown in the

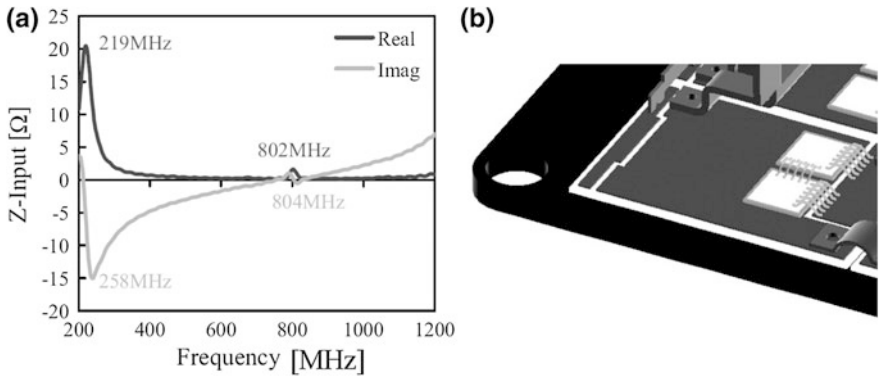


Fig. 14.19 Countermeasure against PETT oscillations: **(a)** Impedance of the module **(b)** detail of the arrangement of the free-wheeling diodes with shorting bond wires. Figure from [Sie06b] © 2006 IEEE

FLO/EMC simulation of this assembly in Fig. 14.19a [Sie06b]. No more resonance point is found in the range of 700 MHz.

The suggestion of [Zim95] is found to be an effective measure against PETT oscillations, in spite of the fact that no knowledge of details of the processes leading to PETT oscillations was available at that time.

To prevent PETT oscillations, the resonance point of the power module must be different from the transit-time frequency f_T governed by the power semiconductor structure. Three-dimensional EMC calculations are useful for this. Ideally, a future simulation system for such tasks should solve the full set of Maxwell equations for the complete construction, and calculate the behavior of the semiconductor devices, e.g. solve the basic semiconductor equations. Although the capabilities of computers steadily increase, this task still seems to be too complex at the moment.

14.3.2 Dynamic Impact-Ionization Transit-Time (IMPATT) Oscillations

IMPATT oscillations have been found as a dynamic oscillatory effect at turn-off of soft-recovery free-wheeling diodes [Lut98]. The attribute “dynamic” is used to indicate that these oscillations occur in connection with a switching event. The dynamic IMPATT oscillation is of high energy, it radiates off disturbances of high intensity and leads to malfunctions of analogue and digital electronic circuits which are present, for example, in drive circuits. A measurement of such an effect is possible in a laboratory setup; such a measurement is shown in Fig. 14.20.

The measurement in Fig. 14.20 was executed in an application in the form of a step-down circuit according to Fig. 5.19. The temperature of the diode was 0 °C. The course of the voltage is plotted inversely. The rated static blocking capability of

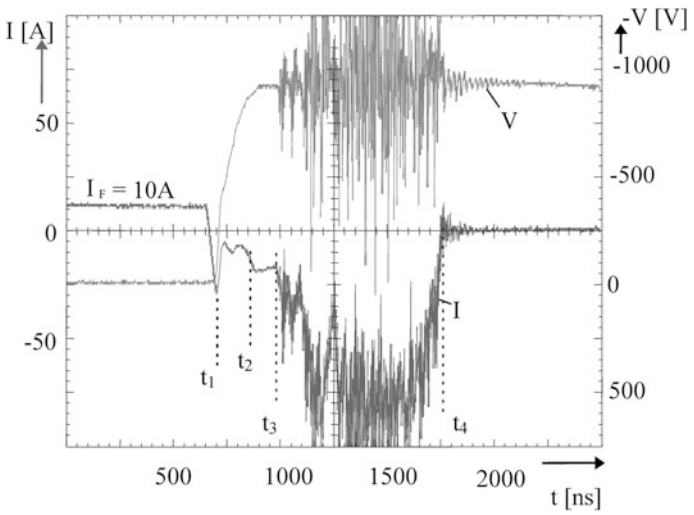


Fig. 14.20 Dynamic IMPATT oscillation of a free-wheeling diode with radiation-induced recombination centers. T = 0 °C. Reprinted from [Lut98] with permission from Elsevier

this diode is 1200 V, its avalanche breakdown voltage is > 1300 V. After the reverse-recovery current peak (time point $t1$) the reverse current decreases. Between t_1 and t_2, the voltage climbs up close to the value of the applied battery voltage, while a tail current flows in the diode. After t_2, a current hump grows from the tail current. This hump occurs only if the battery voltage is above 910 V. A further increase in the battery voltage to 930 V effects that a high reverse current suddenly shoots up from the current hump whose amplitude is a multiple of the reverse-recovery current peak. A high frequency oscillation is superimposed. After some 100 ns (t_4), the oscillation is finished.

A reduction of the voltage by only one ore two volts, or an increase of the temperature by 1 or 2°, removes the effect.

The mechanism of the dynamic IMPATT oscillation is related to that of the IMPATT diode, a device which is likewise intended as oscillator for microwaves [Sze81]. In IMPATT diodes, the static reverse-blocking capability is exceeded and the device is operated in the avalanche mode. In contrast, the dynamic IMPATT oscillation occurs at a voltage significantly lower than the avalanche breakdown voltage. The dynamic IMPATT oscillation is caused by the K-center, which is created at irradiation of the semiconductor with high-energy particles [Lut98]. The energy levels of the most important centers are shown in Fig. 4.30 in Chap. 4. The energy level of the K-center is located below the middle of the band gap. Recent investigations have found its nature as C_iO_i, a defect consisting of an interstitial carbon and oxygen atom [Niw08]. Its contribution to recombination is low, but its density is higher than that of the density of the OV-centers which determine the recombination in irradiated devices if typical annealing processes are used [Sie06].

The K-center has the characteristics of a temporary donor. At forward conduction, if the device is flooded with free carriers, it is occupied with a hole and it is positively charged. The effective doping then becomes

$$N_{eff} = N_D + N_T^+ \tag{14.10}$$

wherein N_T^+ is the density of positively charged K-centers. After the voltage has changed its polarity, the center is discharged:

$$N_T^+(t) = N_T^+ \cdot e^{-\frac{t}{\tau_{ep}}} \tag{14.11}$$

The time constant τ_{ep} of this discharge process is temperature-dependent; it is short (ca. 100 ns at 400 K) at high operation temperatures; it is in the order of some microseconds at temperatures below 300 K. This results in a temporarily increased doping of the device. The doping determines the onset voltage of avalanche breakdown V_{BD}; Eq. (3.84) can be used for the given situation. The increased doping N_{eff} leads to a strongly reduced value of V_{BD}. If now a fast-switching transistor such as an IGBT applies the battery voltage to the diode in a very short time after the current zero-crossing point of the diode, it finds a device with reduced avalanche voltage, and impact ionization sets on. In Fig. 14.20, the temporarily reduced voltage V_{BD} is reached at the time point t_2, and a current created by avalanche occurs as a hump in the current curve. If the device is driven into the avalanche mode even more, the dynamic IMPATT oscillation starts.

The situation in the device during this process is shown in Fig. 14.21. Between t_1 and t_2 a small tail current flows (see Fig. 14.20); in this state, a charge carrier hill from the remaining plasma still exists close to the cathode, and the shape of the electric field is triangular. Electron packets generated by impact ionization run through the electric field to the right-hand side.

A high-frequency AC voltage $V_{RF}.\sin\omega t$ shall be assumed as superimposed to the DC-link voltage V_{DC}, see Fig. 14.22a. It generates a current pulse j_{inj}. Since the

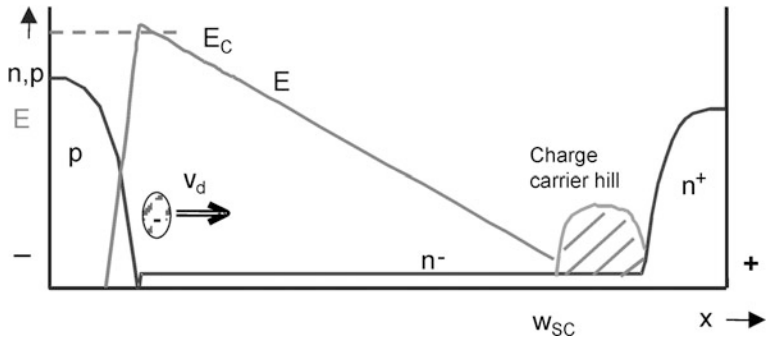


Fig. 14.21 Process in the device at a dynamic IMPATT oscillation

process of impact ionization needs time, the created current pulse is shifted by $\omega t = \pi/2$ to the peak value of the voltage and appears at $\omega t = \pi$, where the AC component of the voltage has its zero-crossing point (Fig. 14.22b). The injected current runs through the space charge at a velocity v_d. Figure 14.22c shows the corresponding current j_{inf} at the terminals of the device according to the Ramo-Shockley-theorem [Eis98]. The current appears at the terminals during the time interval ωt_T which starts at $\omega t = \pi$.

For the generated RF power P_{RF}, Eq. (14.7) holds. P_{RF} is of negative sign and is maximal for $\omega t_T = \pi$, it decreases again for $\omega t_T > \pi$. The diode acts as current source. It emits RF power. For the maximum of generated RF power at $\omega t_T = \pi$, one yields for the frequency of transit-time oscillation at an IMPATT effect [Sze81]

$$f_T = \frac{v_{sat}}{2 \cdot w_{sc}} \tag{14.12}$$

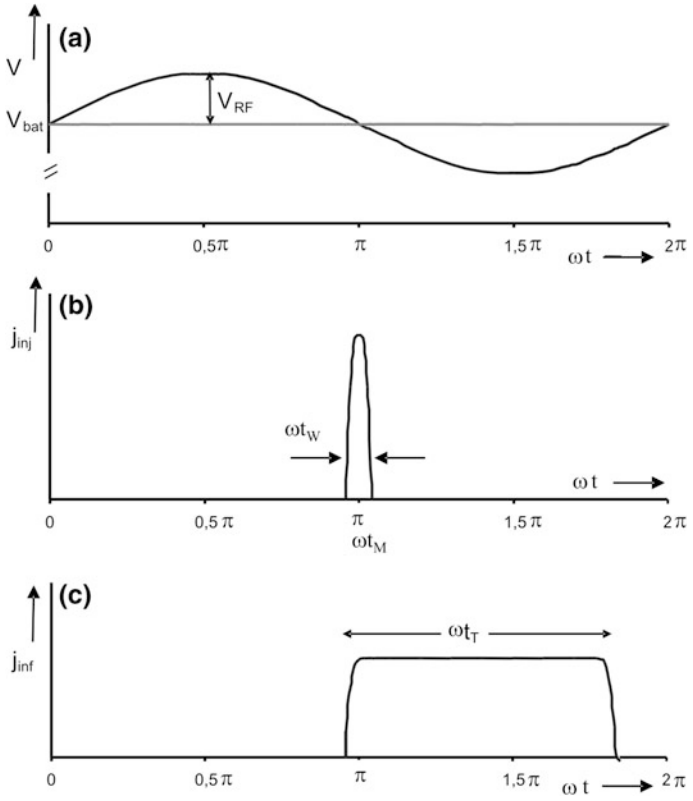


Fig. 14.22 Origin of IMPATT oscillations: **(a)** High-frequency AC voltage superimposed to the applied DC voltage **(b)** Injected current at the time point $\omega t = \pi$. **(c)** Current at device terminals

In contrast to oscillations according to the BARITT effect, there is no phase interval in which the emitted power is damped for the IMPATT effect. The IMPATT effect leads to a high-frequency oscillation of high energy. The RF efficiency of IMPATT diodes can amount up to 30%, as reported in the literature [Eis98]. Additionally, the signal of the IMPATT effect is not in a very narrow frequency band, as depicted in Fig. 14.16 for PETT oscillations, but it contains a lot of noise. This shows obviously that dynamic IMPATT oscillations must be avoided in any case.

IMPATT oscillations can occur if a power semiconductor contains too much K-centers. This can be due to too high electron irradiation doses [Lut98] or high doses of He irradiation projected to an unsuited position in the device [Sie04, Niw08]. For the case of electron irradiation which creates a homogeneous lifetime, a dimensioning rule is given in [Lut98]: Only so many K-centers are allowed that the device can sustain the maximal occurring DC-link voltage in the application (commonly 75% of the rated voltage) at its lowest operation temperature (typ. −40 °C) without occurrence of avalanche. The generation rates of K-centers at irradiation with electrons are given in the literature [Sie06]. If such a rule is observed, the dynamic IMPATT oscillation can surely be avoided.

14.3.3 Transient-Avalanche (TA) Oscillations

A further type of high frequency oscillation was found during the turn-off of 3.3 kV IGBTs at high current. Measurements and simulations indicate that the avalanche generation and transit time effect of carriers within the IGBT leads to this oscillation. The transition time effect takes place during the rise of the collector voltage at turn-off, especially during the dynamic avalanche phase. The range of frequencies is at several 100 MHz. As the oscillation was found at switching with dynamic avalanche, it was named Transient Avalanche Oscillation (TA-Oscillation).

Figure 14.23 shows the measurement of TA oscillations; they are visible in the voltage signal of the wire used as antenna as well as in the waveform of the gate voltage. The oscillations in the gate wiring are a stray-in effect. The oscillations happen at high voltages as the voltage exceeds 2800 V which is the onset of the dynamic avalanche. A large collector current I_C is still flowing and the dV_C/dt is limited by the avalanche process. Oscillations can be found at the beginning, during and at the end of the dynamic avalanche process.

Section 13.4 describes the dynamic avalanche, and for the given situation Eq. (13.13b) can be used

$$V_{av,dyn} = \frac{1}{2} \cdot \left(\frac{8}{B}\right)^{\frac{1}{4}} \cdot \left(\frac{q \cdot N_{eff}}{\varepsilon}\right)^{-\frac{3}{4}} \cdot \left(\frac{\mu_n}{\mu_n + \mu_p}\right)^{\frac{1}{4}} \tag{14.13}$$

The processes in the IGBT at this instant have similarities to the processes described for diodes in Sect. 13.4.2. In a part of the IGBT-base layer the electric field is built

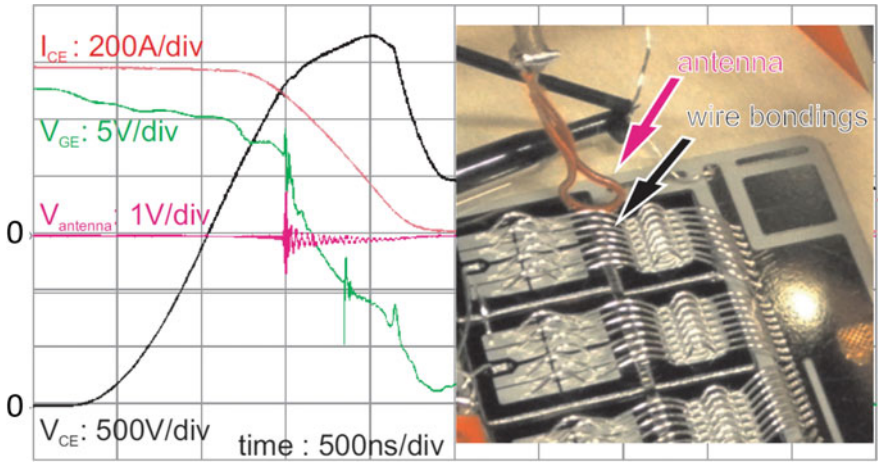


Fig. 14.23 Measurement of TA oscillations at turn-off of 3.3 kV IGBTs. Oscillations visible in an antenna signal as well in the waveform of the gate voltage. Figure from [Hon15]

up, and the current flowing through is carried by holes. This leads to a dynamic avalanche onset well below the rated voltage, see also Fig. 13.12.

The simplified carrier injection mechanism in Fig. 14.22 can be used to calculate the efficiency of RF-power generation in such a transit-time structure. The terminal voltage consists of a DC-part V_{bat} and HF-part V_{RF}. Carriers are injected into the depletion region at ωt_M during the injection angle ωt_W. They need a transit time t_T to drift through the depletion region. The drift angle is ωt_T. As derived from [Eis98], for the RF-efficiency holds, see Fig. 14.22.

$$\eta = \frac{P_{RF}}{P_{DC}} = \frac{-\frac{A}{2\pi}\int_{\omega t_M}^{\omega t_M + \omega t_T} j_{\inf}(\omega t)\cdot V_{RF}\sin(\omega t)d\omega t}{\frac{A}{2\pi}\int_{\omega t_M}^{\omega t_M + \omega t_T} j_{\inf}(\omega t)\cdot V_{DC}d\omega t} \tag{14.14}$$

The solution of (14.15), if additionally the injection angle ωt_W is considered, is

$$\eta = \frac{V_{RF}}{V_{DC}}\cdot\frac{\sin(\omega t_W/2)}{\omega t_W/2}\cdot\frac{\cos(\omega t_M + \omega t_T) - \cos\,\omega t_M}{\omega t_T} \tag{14.15}$$

The three terms in (14.15) are abbreviated as η_1, η_2, η_3. The second term η_2 is for a short $\omega t_W < \pi/5$ close to 1, and for the third term η_3, the transit time dependent part of IMPATT and PETT structures, the solution is drawn in Fig. 14.24. Although the peak values of η_3 decrease with the increase of the drift angle, the second peak value of IMPATT is still as high as the first peak value of PETT. For a certain width of the depletion region w_{sc} each peak value of η_3 in Fig. 14.24 corresponds to a transit time eigenfrequency (f_{1IM}, f_{2IM}, f_{1PE}). Signals at these frequencies would be amplified most effectively.

The resulting frequencies f_{1IM}, f_{2IM}, f_{1BA} can be gained with

$$f_T = \frac{\omega t_T \cdot v_{sat}}{2\pi \cdot w_{sc}} \tag{14.16}$$

As integral effective drift velocity for electrons (IMPATT) $v_{sat,n} = 8.93 \times 10^6$cm/s and for holes (PETT) $v_{sat,p} = 6.44 \times 10^6$cm/s were used. Figure 14.24 shows the dependency of f_{1IM}, f_{2IM}, f_{1BA} on the width of the space charge. For f_{1IM} the result is similar as Eq. (14.12), for f_{1PE} the result is similar to Eq. (14.9). Additionally, f_{2IM} is shown as first harmonics of f_{2IM}.

The TA oscillation usually needs a resonance circuit which is formed by the junction capacity of the devices C_{IGBT} and C_{diode}; further parasitic capacities of the substrate and the wiring close to the dies, see Fig. 14.26a.

Three possible resonance circuits can be identified in Fig. 14.24. The junction capacity of both devices is voltage dependent according to

$$C_j = \frac{\varepsilon \cdot A}{w_{sc}} \tag{14.17}$$

wherein the width of the space charge w_{sc} is voltage dependent, compare Chap. 3.1 Eq. (3.19). The extension of the depletion region leads to the reduction of the chip capacitance both in the IGBT and the diode, which means an increase of the resonance frequencies in all the resonance circuits. Due to the turn-off of the IGBT with high plasma density, the extension of the depletion region of the IGBT is slower than that of the diode. The resonance frequencies shown in Fig. 14.24 are roughly determined with the assumption of 2 times higher w_{sc} for the same voltage

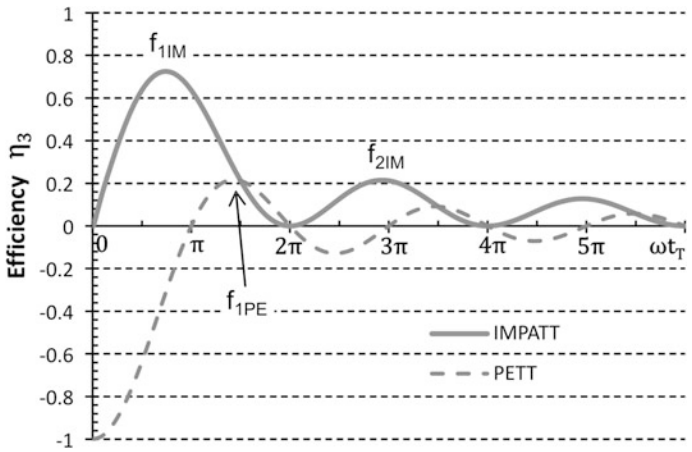


Fig. 14.24 Efficiency η_3 of IMPATT and PETT as function of ωt_T. Figure adapted from [Hon14, Hon15]

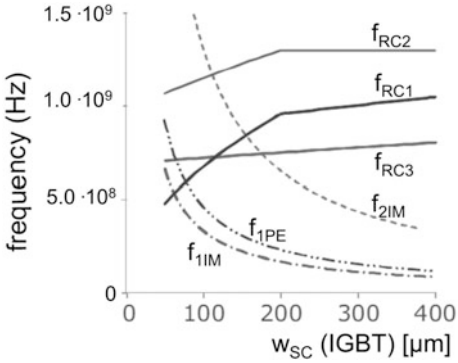


Fig. 14.25 Eigenfrequencies of the depletion region: f_{1IM}, f_{2IM}, f_{1BA} (see Fig. 14.24); eigenfrequencies of resonance circuits (s. Figure 14.26): f_{RC1}, f_{RC2} and f_{RC3}. © 2014 IEEE Reprinted, with permission, from [Hon14]

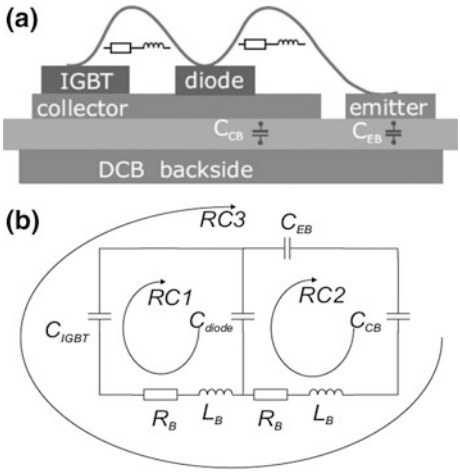


Fig. 14.26 **(a)** Substrate with direct wire bonding between IGBT and diode **(b)** Resonance circuits (RC1, RC2 and RC3) of the setup in a). © 2014 IEEE Reprinted, with permission, from [Hon14]

in the diode, and the depletion region reaches the field stop layer at 400 µm in the considered 3.3 kV diode. Oscillations would be raised under conditions close to the cross points in Fig. 14.25.

Mismatching the resonance circuit and the transit time frequencies can be applied as a countermeasure to the TA-Oscillation. Apart from this, TA-Oscillation can be eliminated by a suitable driver concept: Shortly before the avalanche, an additional short driver pulse is applied to charge the gate and open it again. Partial opening of the MOS-channel can be achieved, delivering an electron current from the MOS-channel. The hole current and hole density at emitter side can thereby be reduced. The corresponding reduction of the peak of the electric field at the blocking junction would suppress the avalanche generation.

14.3.4 Summarizing Remarks on Transit-Time Oscillations

Some aspects of the discussed transit-time oscillations are summarized in Table 14.2. Regarding intensity, the dynamic IMPATT oscillation performs worst, followed by PETT and TA oscillations. However, TA-oscillations are most difficult to avoid completely since the dynamic avalanche occurs at high voltage bipolar Si devices. In cases found up to now, their intensity is low and a resonance circuit is required. However, the IMPATT effect has a high RF efficiency and it cannot be ruled out that under special circumstances, e.g. a strong dynamic avalanche, they even occur without an external resonance circuit with the same resonance frequency.

Even a further type of transit time oscillations, the so-called TRApped Plasma Avalanche Triggered Transit (TRAPATT) oscillation is possible with power devices. It is explained as occurring during avalanche in a device with trapezoidal electric field shape (PT-Design). At a current high enough in the avalanche mode, a plasma of electrons and holes can be formed [DeL70] and an avalanche zone is running through the device.

The RF efficiency of TRAPATT oscillations is found to be very high, even 75% have been obtained [Sze8], which is far above the IMPATT oscillations. In [Kas95] TRAPATT oscillations are explained with numerical simulation of a diode with a

Table 14.2 Summarizing comparison of transit-time oscillations

	Dynamic Impact Ionization transit time (Impatt) Oszillations	Plasma Extraction Transit Time (PETT) Oszillations	Transient Avalanche (TA) Oscillations
Amplitude	200–400 V	<60 V	10 V …
Intensity	Strong radiation. Drivers etc. stop function	Easy overseen at qualification, however disturbs EMC	Hard to find at qualification, however disturbs EMC
Occurrence	Above a defined threshold voltage, which decreases with temperature	May occur at a certain voltage and temperature, can be avoided when changing the conditions	At very special conditions, hard to predict
LC resonance circuit involved?	Not necessary	Yes, necessary	Yes, in most cases
Root cause	Positively charged deep K-centers decrease avalanche onset voltage	Remaining plasma layer injects carriers discontinuously	Mainly dynamic avalanche, PETT mechanism involved
Countermeasure	Device design, limit amount of deep K-centers	Detune LC resonance circuit	- Detune resonance circuit - Short-time electron injection by channel

narrow base at snap-off of the reverse current. The high *di/dt* at snap-off generates a voltage peak larger than the breakdown voltage, which is clamped by the avalanche of the diode. The shown amplitude of the voltage oscillation is almost as high as the applied DC voltage. However, this has occurred in simulations. Snappy diodes are ruled out in most power-electronic applications, see Sect. 14.2.2. Soft recovery behavior is necessary.

Investigations of superjunction MOSFETs with the transmission line pulse (TLP) method in [Chi16] report TRAPATT oscillations at the boarder of the avalanche capability of the device. With the TLP method, short-time high voltage high-current pulses in the avalanche mode are applied, and high-frequency oscillations were found in a short time interval.

However, also at the turn-off of IGBTs voltage peaks are generated which can overshoot the avalanche breakdown voltage. Therefore, it is not ruled out that further oscillation effects will be found.

References

[Bec16] Beckedahl, P., Buetow, S., Maul, A., Roeblitz, M., Spang, M.: 400 A, 1200 V SiC power module with 1nH commutation inductance. In: Proceedings CIPS 2016, VDE Verlag GmbH (2016)

[Chi16] Chirilă, T., Reimann, T., Rüb, M.: Dynamic avalanche in charge-compensation MOSFETs analyzed with the novel single pulse EMMI-TLP method. In: Proceedings of 2016 IEEE International Reliability Physics Symposium (IRPS), pp. 1–5 (2016)

[DeL70] DeLoach B.C., Scharfetter D.L.: Device physics of TRAPATT oscillators. IEEE Trans. Electron Devices **17**(1), 9–21 (1970)

[DIN00] DIN EN 55011 – Industrielle, wissenschaftliche und medizinische Hochfrequenzgeräte; Funkstörungen – Grenzwerte und Messverfahren, VDE-Verlag GmbH, Berlin (2000)

[Eis98] Eisele, H., Haddad, G.: Active microwave diodes. In: Sze, B.M. (eds.) Modern Semiconductor Device Physics. Wiley, New York (1998)

[Eld98] El-Dwaik, F.: Ein Beitrag zur Optimierung des Wirkungsgrades und der EMV von Wechselrichtern für batteriegespeiste Antriebssysteme, Dissertation, Chemnitz (1998)

[Fei15] Feix, G., Hoene, E., Zeiter, O., Pedersen, K.: Embedded very fast switching module for SiC Power MOSFETs. In: Proceedings PCIM Europe, pp. 104–110 (2015)

[Fuj12] Shigeto Fujita, S.: Simulation study on insulated gate bipolar transistor turn-off oscillations. Jpn J Appl Phys **51**, 054101 (2012)

[FLO04] Flomerics Ltd.: FLO/EMC Reference Manual Release 1.3, 2004

[Gut01] Gutsmann, B., Silber, D., Mourick, P.: Explanation of IGBT tail current oscillations by a novel plasma extraction transit time mechanism. In: Proceeding of the 31st European Solid-State Device Research Conference, pp. 255–258 (2001)

[Gut02] Gutsmann, B., Mourick, P., Silber, D.: Plasma extraction transit time oscillations in bipolar power devices. Solid-State Electron. **46**(5), 133–138 (2002)

[Hon14] Hong, T., Pfirsch, F., Bayerer, R., Lutz, J., Silber, D.: Transient avalanche oscillation of IGBTs under high current. In: Proceedings ISPSD (2014)

[Hon15] Hong, T.: Transient avalanche oscillation of IGBTs under high current, Ph.D. thesis Chemnitz (2015)

[Kas95] Kaschani, K.T., Sittig, R.: How to avoid TRAPATT oscillations at the reverse recovery of power diodes. In: International Semiconductor Conference, CAS'95 Proceedings, pp. 571–574 (1995)

[Kas97] Kaschani, K.T.: Untersuchung und Optimierung von Leistungsdioden, Dissertation, Braunschweig (1997)

[Lut98] Lutz, J., Südkamp, W., Gerlach, W.: IMPATT oscillations in fast recovery diodes due to temporarily charged radiation induced deep levels. Solid-State Electr. **42**(6), 931–938 (1998)

[Niw08] Niwa, F., Misumi, T., Yamazaki, S., Sugiyama, T., Kanata, T., Nishiwaki, K.: A Study of correlation between CiOi defects and dynamic avalanche phenomenon of PiN diode using he ion irradiation. In: Proceedings of the PESC, Rhodos (2008)

[Omu030] Omura, I., et al.: Electrical and mechanical package design for 4.5 kV ultra high power IEGT with 6kA Turn-off capability. In: Proceedings of the ISPSD, Cambridge (2003)

[Pal99] Palmer, P.R., Joyce, J.C.: Causes of parasitic current oscillation in IGBT modules during turn-off. In: Proceedings of the EPE, Lausanne (1999)

[Sie03] Siemieniec, R., Lutz, J., Netzel, M., Mourick, P.: Transit time oscillations as a source of EMC problems in bipolar power devices. In: Proceedings of the EPE, Toulouse (2003)

[Sie04] Siemieniec, R., Lutz, J., Herzer, R.: Analysis of dynamic impatt oscillations caused by radiation induced deep centres with local and homogenous vertical distribution. In: IEEE Proceedings Circuits, Devices and Systems, vol. 151(3), pp. 219–224 (2004)

[Sie06] Siemieniec, R., Niedernostheide, F.J., Schulze, H.J., Südkamp, W., Kellner-Werdehausen, U., Lutz, J.: Irradiation-induced deep levels in silicon for power device tailoring. J. Electrochem.l Soc. **153**(2) G108-G118 (2006)

[Sie06b] Siemieniec, R., Mourick, P., Netzel, M., Lutz, J.: The plasma extraction transit-time oscillation in bipolar power devices – mechanism, EMC effects and prevention. IEEE Trans. El. Dev. **53**(2) 369–379 (2006)

[Sze81] Sze, S.M.: Physics of semiconductor Devices. Wiley, New York (1981)

[Win12] Wintrich, A.: "Herausforderungen beim Einsatz von SiC in Hochleistungsmodulen", ELEKTRONIKPRAXIS Leistungselektronik & Stromversorgung, Mai (2012)

[Zim95] Zimmermann, W.: Sommer KH, Patent DE 19549011C2 (1995)